

PolyAgora Meta-Regime Steering Layer (V0.1)

CTO Technical Spec — Discussion Consolidation

1. Strategic Objective

The next architectural phase of PolyAgora is not to replace existing strategies (e.g. Momentum), but to introduce a higher-order runtime steering layer capable of dynamically allocating among multiple admissible strategy manifolds.

The objective shifts from:

- relative benchmark outperformance

toward:

- attractive absolute returns
- recoverability
- maneuverability
- institutional-grade robustness
- low correlation to traditional asset classes

in line with AGUR's framing:

- Sharpe > 1.5
 - Sortino > 1.5
 - Calmar > 1.5
 - low equity/bond correlation
 - robust live-trading survivability.
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2. Core Concept

We construct:

Level 1

Existing strategy manifolds:

- PolyAgora V7
- Momentum
- Defensive / CHASH

Each remains independently defined and frozen initially.

Level 2

A higher-order MRTP-driven steering layer dynamically allocates among those manifolds.

This is NOT:

- "Momentum vs PolyAgora"

but:

- PolyAgora governing admissibility of strategy geometries.
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3. Conceptual Interpretation

Each strategy becomes:

- a regime-space
- a maneuverability geometry
- a recoverability profile

Examples:

Momentum

Strong under:

- persistent trends
- coherent low-vol expansions

Weak under:

- violent reversals
 - fragmentation
 - transition shocks
-

PolyAgora V7

Strong under:

- transitions
 - fragmentation
 - boundary instability
 - recovery phases
 - regime deformation
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Defensive / CHASH

Strong under:

- rupture
 - inadmissibility
 - collapse geometry
-

4. Layer Separation Principle

CRITICAL:

The meta-layer must remain:

- slower

- coarser
- lower entropy
- governance-oriented

than underlying manifolds.

The system MUST avoid:

- recursive steering instability
 - oscillatory reflexivity
 - self-referential feedback loops
 - hyper-reactive switching
-

5. Minimal Architecture

Asset Layer

Existing:

- assets
- signals
- returns
- weights

No change.

Strategy Manifold Layer

Each manifold outputs:

- asset weights
- expected returns
- volatility profile
- drawdown profile
- turnover
- admissibility diagnostics

Examples:

M_1 = PolyAgora V7

M_2 = Momentum

M_3 = Defensive

Meta Steering Layer

Inputs:

- regime persistence
- volatility stress
- boundary tension
- recoverability
- admissibility confidence

- transition density

Outputs:

- manifold allocations

6. Meta State Definition

Define:

(r_t, v_t, c_t, b_t)
]

Where:

- (r_t) = regime persistence
- (v_t) = volatility stress
- (c_t) = admissibility confidence
- (b_t) = boundary tension

Sources may include:

- VIX
- survival matrix entropy
- zone transitions
- drawdown acceleration
- realized volatility
- VAIDM / ADD overlays

7. Strategy Admissibility

Each manifold receives:

[
 $A_i(t) \in [0,1]$
]

Meaning:

"How admissible is this strategy manifold under current geometry?"

8. MRTP Meta Steering

For each manifold:

$\gamma D_i(t)$
]

Where:

- (S_i) = persistence stability
- (F_i) = future maneuverability

- (D_i) = expected damage if wrong

Interpretation:

M RTP does NOT maximize immediate return.

M RTP maximizes:

- future reachable admissible states
- recoverability
- maneuverability preservation

9. Meta Allocation

Define manifold allocation:

$$\frac{A_i(t) \exp(\lambda \text{M RTP}_i(t))}{\sum_j A_j(t) \exp(\lambda \text{M RTP}_j(t))}$$

Where:

- (W_i) = allocation to manifold i
- λ = aggressiveness parameter

This produces:

- smooth transitions
- probabilistic steering
- no hard switching

10. Final Portfolio

Each manifold produces:

$$w_i^{\{\text{assets}\}}(t)$$

Final portfolio:

$$\sum_i W_i(t) \cdot w_i^{\{\text{assets}\}}(t)$$

11. Runtime Timing

Recompute:

- daily

- 16:00 ET close

Using:

- forward-safe information only
- no same-bar contamination
- no hindsight

Consistent with AGUR conventions.

12. Validation Protocol

CRITICAL.

The meta-layer MUST NOT see future returns.

At time (t), it only sees:

[
 $\mathcal{F}_{t-1}$
]

Meaning:

- realized information only
 - no future manifold performance
-

Validation Steps

A. Freeze Manifolds

Initial versions remain fixed:

- V7
- Momentum
- Defensive

No online mutation.

B. Walk-Forward Validation

Example:

- Train: 2015–2019
- Test: 2020

Then roll forward.

NO global optimization.

C. Crisis Validation

Test specifically:

- COVID
- inflation transitions
- volatility spikes
- trend crashes
- reversal regimes

This architecture is fundamentally about:

- transition maneuverability.
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D. Meta Advantage Validation

Compare:

1. Static Momentum
2. Static PolyAgora
3. Meta-Steered System

Measure:

- Sharpe
 - Sortino
 - Calmar
 - drawdown
 - recoverability
 - turnover
 - switching frequency
 - correlation profile
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13. Meta-Level Fixed Star / Survival Geometry

YES:

The meta-layer itself requires:

- survival structure
- admissibility topology
- higher-order Fixed Stars

But now:

the objects are not assets.

The objects are:

- strategy manifolds.
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Meta Survival Matrix

Define:

$$[S^{\{\text{meta}\}_{ij}}]$$

Meaning:

- admissibility/survival relation between strategy manifolds.

Examples:

- Momentum survives persistent expansions.
- PolyAgora survives fragmentation.
- Defensive survives collapse.

This may create:

- local stars
- manifold blocks
- strategic geometries

14. Important Distinction

Meta Fixed Star is NOT:

- "best performing strategy."

It is:

- stable maneuverability center under current geometry.

This distinction is essential to avoid:

- hindsight optimization
- performance chasing
- hidden overfit

15. Reference Polygon (RP)

Initial Recommendation

Use SAME RP for:

- asset level
- meta level

Reason:

- coherence
- semantic alignment
- stable admissibility structure

Long-Term Possibility

Eventually:

meta layer may evolve toward a more abstract RP describing:

- persistence
- recoverability
- convexity
- fragmentation
- maneuverability

while asset layer RP continues describing:

- market geometry
- sector structure
- macro relations

DO NOT separate yet.

16. Critical Constraints

DO NOT:

- recursively steer steering layers
 - add meta-meta learning
 - add adaptive manifold mutation
 - create high-frequency switching
 - create hyper-reactive transitions
-

17. Initial Implementation Recommendation

INITIAL VERSION SHOULD ONLY USE:

- PolyAgora V7
- Momentum
- Defensive / CHASH

Do NOT yet add:

- carry
- convexity overlays
- macro engines
- online learning

First validate:

- manifold steering stability
 - recoverability advantage
 - admissibility quality
 - transition handling
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18. Strategic Interpretation

PolyAgora is evolving from:

- a strategy engine

toward:

- a runtime governance architecture over strategy spaces.

This is the conceptual shift.

The system no longer merely optimizes positions.

It governs:

- admissibility
- maneuverability
- recoverability
- strategy transitions

under changing market geometries.

END